United International University

B.Sc. in Computer Science & Engineering (CSE)

CSE 4889: Machine Learning

Midterm Exam: Summer 2025 Time: 1 Hour 30 Minutes Marks: 30

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Answer all of the following questions.

1. a) Suppose you train a sentiment analysis model using Facebook posts as the training data and then evaluate it on movie reviews from the Internet Movie Database (IMDb). Would this be considered an open test or a closed test? Justify your answer with a clear explanation. [3+3]

   b) UIU plans to analyze the research interests of 100 students based on a similarity score (ranging from 0 to 1) computed from their publication keywords. The students naturally form overlapping interest groups (e.g., AI & ML, Cybersecurity, Data Science, Networks, Robotics), but the exact number of groups is unknown. UIU is also interested in exploring relationships between groups (e.g., AI is more closely related to Data Science than to Networks). Given that the dataset is relatively small (100 students), computational cost is not a concern. Between k-means clustering and hierarchical clustering, which method would you recommend for this task? Justify your choice.

2. You are given the following set of 2D data points: [2+4+2]

   A1(1, 3), A2(2, 4), A3(3, 5), A4(7, 8), B1(1, 2), and B2(4, 5).

   Using an appropriate clustering approach and distance metric:

   a) Construct the pairwise distance matrix for the data points.
   b) Draw the resulting Dendrogram to illustrate the merging process.
   c) Based on the Dendrogram, suggest the most reasonable number of clusters for this dataset and justify your choice.

3. Consider the following 2D data points. Cluster the data points using a precise density-based clustering method where k=2 (k denotes the nearest neighbor). [8]

Table: 2

x	y	Data points
1	2	A
2	2	B
2	3	C
7	8	D
8	7	E
25	80	F

4. A dataset of products sold in different shops is given as follows: [8]

Product	Shop	Prices	VAT	Discount
Doughnut	Tesco	750	81	6.5
Doughnut	Tesco	800	86	7.5
Dark chocolate	Unimart	250	87	8.0
Dark chocolate	Unimart	675	98	9.0
Ice cream	Asda	3000	120	10
Ice cream	Asda	750	115	9.5

You are asked to train a Random Forest Classifier (with resampling where each tree uses at least 3 samples, i.e., $S \geq 3$) to classify products based on the features Shop, Prices, VAT, and Discount**.**

After training, classify the following unknown samples:

Product	Shop	Prices	VAT	Discount
?	Tesco	325.5	46.5	3.5
?	Unimart	1800	120	12.5
?	Asda	500	50	5.0

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